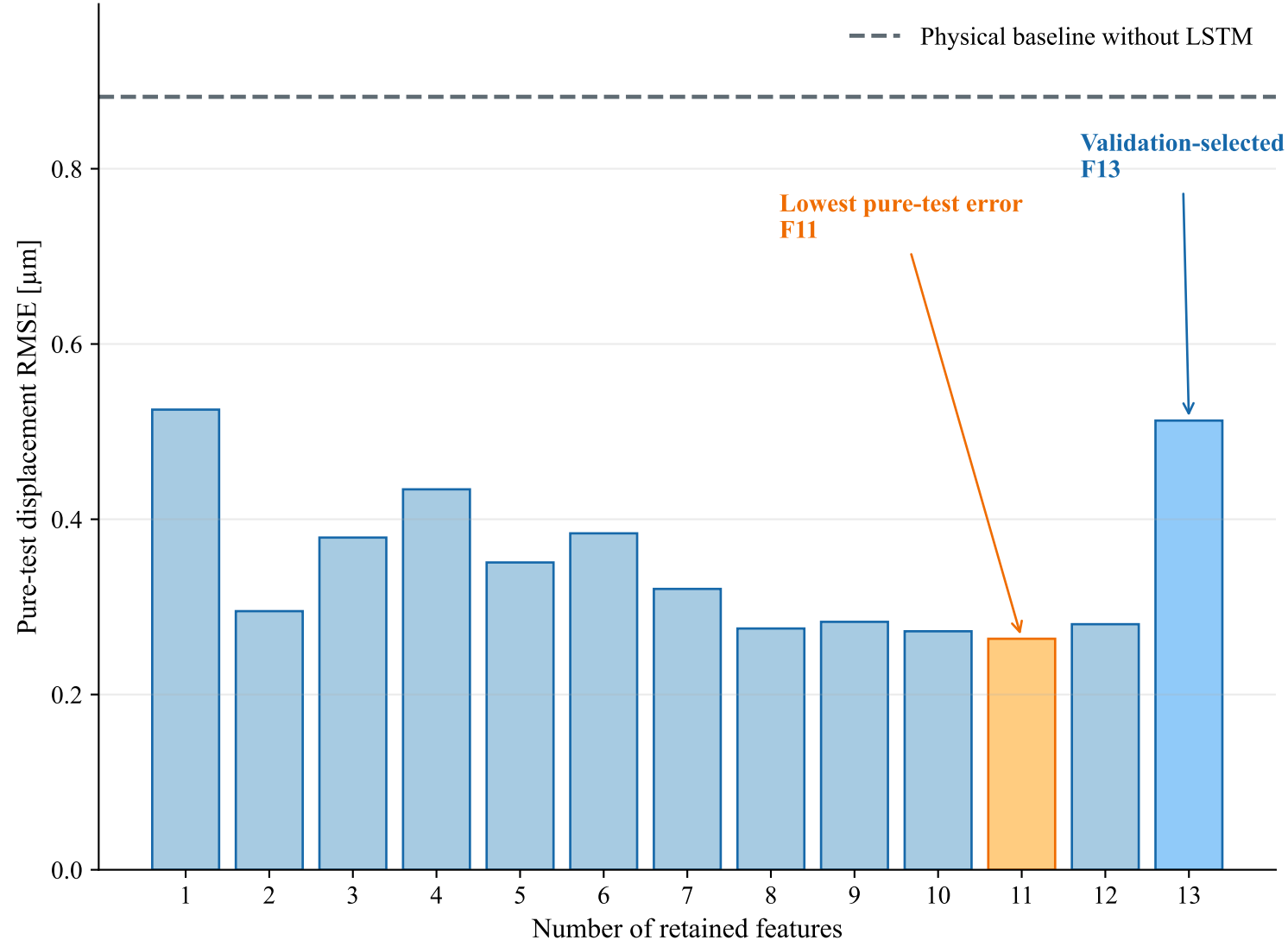


Interpretation of the Cumulative Feature-Ablation Study

Purpose: quantify feature sensitivity, identify influential or redundant inputs, and assess whether a simpler model can preserve predictive accuracy.

Pure-Test Displacement Error Across Feature Configurations



Principal Observations

1. Distinct model responses

Reduced-feature predictions differ from F13 by as much as 0.0045538 mm.

2. Dominant accuracy transition

The largest validation improvement occurs when previous acceleration is introduced in F10.

3. Validation and pure-test behavior

F13 minimizes validation RMSE, whereas F11 minimizes pure-test RMSE.

4. Pure-test accuracy

The lowest worst-case displacement fit is 99.04% across all configurations.

5. Statistical limitation

The present comparison uses one seed; repeated-seed analysis is required to quantify uncertainty.

Scope: F01 uses current as the sole LSTM correction feature; the hybrid one-step predictor continues to use the displacement-history physical baseline.